MAHINDRA UNIVERSITY, HYDERABAD

École Centrale School of Engineering

Minor-2 Examination

Program: B.Tech Branch: CAM Year:III

High-Performance Computing (MA3102)

Date:29-10-2025
Duration: 90 minutes

Start Time: 10:00AM
Max. Marks: 60

Instructions:

1. Open book examination. Students will be allowed to the exam with the textbook: "The Art of Multiprocessor Programming" by Maurice Herlihy and Nir Shavit, Morgan Kaufmann Publishers.
2. Answer all questions.

---

1. Explain how parallelism, scalability, and communication overhead influence the performance of a High-Performance Computing (HPC) system. Illustrate your answer with an example and discuss how its performance might change as the number of processors increases.

   (CO1, CO3)(20 Marks)

2. Design an $n - thread$ lock using $2 - thread$ Peterson locks as building blocks. Describe in detail the architecture and functioning of your design, ensuring that it adheres to the mutual exclusion property. Verify whether your design ensures mutual exclusion for all n threads. Provide a concise proof sketch to substantiate your claim.

   (CO2)(20 Marks)

3. Design and implement algorithms for the *contains()*, *insert()*, and *remove()* operations on the concurrent binary search tree. The operations should ensure thread safety and optimal performance in multi-threaded environments. Also, describe how your algorithms maintain the correctness.

   (CO2, CO3)(20 Marks)